IEOR 173 Lecture Notes

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Chapter 1

1.1 Lecture 03

1.1.1 Joint Distributions

$$P(x, y) = P(X \leq x, y \leq y)$$

This is a cumulative distribution for joint distributions.

If $X \perp Y$ (independent), (perpendicular), thee the join distribution is the product of the marginals. $P(X, Y) = P(X)P(Y) \forall X, Y$

1.1.2 Cov + Correlations

Covariance is a measure of linear dependence.

$$Cov(X, Y) = E[(x - EX)(Y - EY)] = EXY - EXEY$$

Interpretation: If they vary together, Cov is larger.

Properties

1. Cov(X,X) = Var(x)
2. Cov(X,Y) = Cov(Y,X)
3. Cov(cX, Y) = cCov(X,Y)
4. Cov(X, Y+z) = Cov(X,Y) + Cov(X,Z)

Correlation:

$$p(X, Y) = \frac{Cov(X, Y)}{sd(X) \cdot sd(Y)}$$

If

- ¿ 0, pos correlation
- ¡ 0, negative correlation
- = 0 No correlation

Note:-

Independent does imply uncorrelated, however uncorrelated does not imply independent.
Ex: If $P(-1, 1) = P(0, 0) = P(1, 1) = \frac{q}{3}$. then $E[XY] = \frac{0}{3} = 0$. However, $E[X] = 0, EY =?$, therefore $E[XY] = 0$ and $Cov(X, Y) = 0$. There is no linear relationship, however there is a dependency for $y = x^2$.

Therefore, we can generalize our variance formula

$$Var(A + \sum b_i X_i) = \sum b_i^2 Var(X_i) + \sum_{\forall ij, i \neq j} b_i b_j Cov(X_i, X_j)$$

1.1.3 Hat Example

Example 1.1.1 (Lists of number)

Imagine go to party, men take off hats, women keep on hats. Each man randomly takes a hat and leaves.
Let $X = \#$ of matches in N pairs (man and his hat)
What is EX and Var(X)?
If two men, the only possibility of matches are 0 and 2. The mean is 1. If three men, a similar pattern occurs.

$$EX = 1 \forall N$$

As N grows larger, the probability decreases, but there is more men to balance it out.

$$Var(X) = 1 \forall N \geq 2$$

We can prove this by using Indicators.
Let $X = \sum^N I_i$, where $I_i = I$it is a match $\sim Bern(\frac{1}{N})$

$$EX = E[\sum I] = \sum^N E[I] = N\frac{1}{N} = 1$$

$$Var(I_i) = E[I^2] - [EI]^2 = p - p^2 = p(1-p) = \frac{1}{N} - \frac{1}{N^2} = \frac{N-1}{N^2}$$

$$Cov(I_i, I_j) = E[I_i I_j] = EI_i EI_j$$

$$E[I_i I_j] = P(\text{both i and j match}) = P(I - i = I_j = 1)$$

$$P(I_i = 1)P(I_j = 1 \mid I_i = 1) = \frac{1}{N} \cdot \frac{1}{N-1}$$

Conceptually, we can think of event j following event j

$$Cov(I_i, I_j) = \frac{1}{n(n-1)} - \left(\frac{1}{n}\right)^2 = \frac{1}{N^2(N-1)}$$

$$Var(X) = Var(\sum I_i) = \sum Var(I) + \sum Cov(I_i, I_j) = N \cdot \frac{N+1}{N^2} + N(N-1)\frac{1}{N^2(N-2)} = 1$$

The calculation will be left as an exercise to the future reader.
As $N \to \infty$,

$$Cov(I_i, I_j) \to 0$$

$$Var(I_i) \to \frac{1}{N}$$

$$X \to Bin(N, \frac{1}{N}) \sim Poisson(1)$$

Intuitively, if there are thousands of men there, whether or not 1 person gets his hat is not really going to affect the chances another gets his.

1.1.4 Start condition (Ch 3)

What if we have dependent random variables and want to compute conditional probabilities?

Conditions:

1. discrete: $P(X, y) = P(X = x, X = y), P(x, y) = P(X \leq x, Y \leq y) = \sum_{\leq x} \sum_{\leq y} P(x, y)$
2. continuous: $f(x, y) = F(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v) du dv$

3. marginal: $F_X(x) = P(X \leq x) = F(X, \infty)$
4. conditional: $P_{x|y}(X \mid y) = P(X = x|Y = y) = \frac{P(X=x,Y=y}{P(Y=y)} = \frac{P_{x,y}(X,y)}{P_y(y)} = \frac{\text{joint}}{\text{marginal}}$

Chapter 2

2.1 Lecture 04

2.1.1 Properties of condition expectation

$$E[X|X = x] = x \forall x$$

$$E[X|Y = y]$$

, is a number bu $E[X|Y]$ is a random variable.

$$X \perp W, X \sim exp(\lambda), W \sim exp(\lambda)$$

$$f_x(x) = \lambda e^{-x} \text{for x} \geq 0$$

$Y = X + W$, condition on $Y = y$, $E[X|Y = y]$. We need $f_{x|y}(x|y)$

$$X|y + W = [y - X|y]$$

This should have the same distributions.

To show this, we need to get the distribution function.

$$f_{x|y}(x|y)\frac{f_{x,y}(x,y)}{f_y(y)} = \frac{f_{x,w}(x, y - x}{f_y(y)} = \frac{f_x(x)f_w(y - x)}{f_y(y)} = \lambda e^{-\lambda x}\lambda e^{-\lambda(y-x)} = \lambda^2 e^{-\lambda y}$$

For $0 \leq x \leq y$

We also need to get the marginal distributions.

$$f_y(y) = \lambda^2 \int_{-\infty}^{\infty} f_{x,y}(x,y)dx = \lambda^2 \int_0^y e^{-\lambda y}dx = \lambda^2 y e^{-\lambda y}$$

$$f_{x|y}(x|y) = \frac{f_{x,y}(x,y)}{f_y(y)}\frac{\lambda^2 e^{-\lambda y}}{y\lambda^2 e^{\lambda y}} = \frac{1}{y}$$

This is a uniform distribution from $0 \leq x \leq y$. Therefore $E[X|Y = y] = \int_0^y (x\frac{1}{y})dx = \frac{y}{2}$

Instead, if we take $G(y) = E[X|Y]$, which is a random variable, we can get $E[E[X|Y]$.

We can do this by $E[g(y)] = \int_{-\infty}^{\infty} g(x)D(x)dx$, which is the probability $g(X)$ multiplied by the density $d(x)$.

$$E[E[X|Y]] = \int E[X|Y = y] f_y(y) dy$$

.

It turns out that this becomes $E[X]$.

Example 2.1.1
$Y = \sum^N X_i, EX_i = \mu$, $N \perp X_i$'s, or N is independent of the X_is.

$$E[Y] = E[\sum^N y_n] = E[E[\sum^N X_i|N]] = \sum_{\text{all } n} E[\sum^N X_i|N = n] p_n(n) = \sum_{\text{all} n} E[\sum m^n X_1] p_n(n) = \sum_{\text{all} n} n\mu p_n(n)$$

$$= \mu \sum n p_n(n) = \mu EN$$

Where N is discrete, integer valued, nonnegative. $p_n(n)$ is the pmf for n

Example 2.1.2 (Mean of a geometric)

$$X \sim geom(p)$$

Find EX.
Because geometric dists are memoryless, we can condition on the first step.
Let $I = I\{\text{1st trial is a success}\} \sim Bern(p)$
Compute EX

$$EX = E[E[X|I]] = \sum_{\text{all i}} E[X|I = i] p(I = i) = E[X|I = 1] p(I = 1) + E[X|I = 0] p(I = 0)$$

$$= 1p + (1 + EX)(1 - p) \implies EX = \frac{1}{p}$$

Basically saying, we either get the heads on this trial, or then we have to restart.

Example 2.1.3 (Packet Transmission)
Serve 1 packet per time slot. Each slot we have $A_i = \#$ arrivals in slot i. We assume i.i.d. and has p_0, p_1, p_2.
We assume that the buffer is size 2. (We can only hold 2 packets in the buffer)
Basically, for every time slot, we can only store up to 2 packets. Each timeslot, one packet leaves the buffer, and if there is too many packets, the incoming packet has to bounce.

$$N = \text{length of busy period}$$

A busy is period is where there are arrivals to an empty system and ends when the system is emptied again.
Need $E[N]$
First thing that we need to think about is conditioning. It would be helpful to know how many packets start the busy period, therefore we condition on $X = \#$packets that start the busy period (b.p), where $X \in \{1, 2\}$. Thus we also need to know $P(X = 1) = P(A = 1|A = 1\text{or} = 2) = \frac{p_1}{p_1+p_2}$ and $P(X = 2)i = P(A = 2|\dots) = \frac{p_2}{p_1+p_2}$.

$$E[N] = E[E[N|X]] = EN|X = 1]\frac{p_1}{p_1 + p_2} + E[N|X = 2]\frac{p_2}{p_1 + p_2}$$

Chapter 3

3.1 Lecture 05

3.1.1 Packet Transmission (cont)

Example 3.1.1 (Packet Transmission (cont))

Packet transmission appends A_i i.i.d. p_0, p_1, p_2. N = length of busy period. We still want $E[N]$

$$X = \begin{cases} 1 \frac{p_1}{p_1+p_2} \\ 2 \frac{p_2}{p_1+p_2} \end{cases}$$

$$E[N] = E[E[N|X]] = E[N|X=1]\frac{p_1}{p_1+p_2} + E[N|X=2]\frac{p_2}{p_1+p_2}$$

We now need to compute $E[N|X=n]$.
For simplicity $E[N_1] = E[N|X=1]$:
It would help to know how many packets we get at time 1

$$E[N_1] = E[E[N_1|A]]$$

where A is the number of arrivals in the slot of the busy period.

$$\begin{aligned} E[N] &= E[N_1|A=0]p_0 + E[N_1|A=1]p_1 + E[N_1|A=2]p_2 \\ &= 1p_0 + (1+E[N_1])p_1 + (1+E[N_2])p_2 \\ &. \end{aligned}$$

We can use the $1 + E[N_1]$ because of the i.i.d. and the memory less state.

$$E[N_2] = E[N_2|A=0]p_0 + E[N_2|A=1]p_1 + E[N_2|A=2]p_2$$

$$E[N_2] = (1+E[N_1])p_0 + (1+E[N_2])p_1 + (1+E[N_2])p_2$$

The last term is because the buffer only has storage for 2 packets. If one goes out, then one gets replaced and the other gets sent away.
We can thus solve for $E[N_1]$ and $E[N_2]$, which we can plug into the original equation.
This is a very simple Markov Chain.

3.1.2 Computing Variance:

$\mathrm{Var}(x)$ in two methods

1. Use $\text{Var}(x) = E[X^2] - E[X]^2$, condition for $E[X^2] + E[X]$
2. Condition Var formula

   Example for method 1:

$$X \sim geo(p)$$

$$E[X^2] = E[E[X^2]|I] \text{where} I = \text{indicator of success } \sim \text{Bern}(p)$$

$$\begin{aligned} E[X^2] &= E[X^2|I = 1]p + E[X^2|I = 0](1-p) \\ &= 1p + (E[(X+1)^2])(1-p) = p + E[X^2 + 2x + 1](1-p) \\ &= p + (E[X^2] + E[2X] + 1)(1-p) \end{aligned}$$

   Solve for $E[X^2] = \frac{n-p}{p^2}$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{2-p}{p^2} - \left(\frac{1}{p}\right)^2 = \frac{1-p}{p^2}$$

   How do we get the method 2 formula?

Proof. Conditional Variance Formula

$$\begin{aligned} [E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]) &= E[E[X|Y^2] - (E[X|Y])^2] + E[E[X|Y]^2] - E[E[E[X|Y]^2]]^2 \\ &= E[E[X^2|Y]] - E[E[X|Y]]^2 + E[X^2] - E[X]^2 \\ &= \text{Var}(X) \end{aligned}$$

.

Theorem 3.1.1 $X \perp Y \implies \text{Var}(XY) = \text{Var}(X)$

$$X \perp Y \implies \text{Var}(XY) = E[Y^2]\,\text{Var}(X) + E[X]^2\,\text{Var}(Y) = E[X^2]\,\text{Var}(Y) + E[Y]^2\,\text{Var}(X)$$

$$\text{Var}(XY) = E[\text{Var}(XY|Y)] + \text{Var}(E[XY|Y])$$

.

For the first term:

$$\text{Var}(XY|Y = y) = \text{Var}(Xy|Y = y) = y^2\,\text{Var}(X) \implies \text{Var}(XY|Y) = Y^2\,\text{Var}(X)$$

We can do this because X is independent of Y.

For the second term:

$$E[XY|Y = y] = yE[X] \implies E[XY|Y] = YE[X]$$

Therefore,

$$\text{Var}(XY) = E[Y^2\,\text{Var}(X)] + \text{Var}(YE[X]) = \text{Var}(X)E[Y^2] + E[X]^2\,\text{Var}(Y)$$

Theorem 3.1.2

X_i is i.i.d. N is an integer valued random variable independent of the X_i s. Find the $\mathrm{Var}(\sum^N X_i)$

$$\mathrm{Var}(\sum^N X_i) = E[\mathrm{Var}(\sum^N X_i|N)] + \mathrm{Var}(E[\sum^N X_i|N)]$$

For the first term:

$$\mathrm{Var}(\sum^N X_i|N = n) = \mathrm{Var}(\sum^n X_n|N = n) = \sum^N \mathrm{Var}(X|N = n) = \sum^n \mathrm{Var}(X) = n\,\mathrm{Var}(X_i)$$

We proved the second term earlier.

$$E[\mathrm{Var}(\sum^N X_i)] = E[N\,\mathrm{Var}(X_i)] + \mathrm{Var}(NE[X_i]) = E[N]\,\mathrm{Var}(X_i) + \mathrm{Var}(N)E[X_i]^2$$

If $N \perp X$

$$\mathrm{Var}(NX) = E[X^2]\,\mathrm{Var}(N) + E[N^2]\,\mathrm{Var}(X)$$

This one is like the catering example, where everyone orders the same thing. The other is if all the X_i are i.i.d., therefore ordering different things. Therefore the first is more variable than the second.

Chapter 4

4.1 Lecture 06

4.1.1 Recall:

$$I\{A\} \sim \text{Bern}(P(A)) \implies E[I\{A\}] = P(A)$$

$$EX = E[E[X \mid y]]$$

Therefore,

$$\begin{aligned}
P(A) &= E[I\{A\}] \\
&= E[E[I\{A \mid Y\}]] \\
&= E[P(A \mid y)] \\
&= \sum_y P(A \mid Y = y) p_y(y) \quad \text{or} \int_{-\infty}^{\infty} P(A \mid Y = y) f_y(y) dy \\
&= \text{Total probability rule}
\end{aligned}$$

These concepts are important for Markov chains!

Example 4.1.1

i.i.d. rolls of 2 fair dice. What the chance of sum =5 before sum =7
We could enumerate all the possibilities.

$$P(5\text{on a single trial}) = \frac{4}{36}$$

$$P(7\text{on a single trial} = \frac{6}{36}$$

We can do the computation 2 ways.

$$P(5 \text{ before } 7) = P(E)$$

Conditioning!!

Let X = sum on 1st roll.

$$\begin{aligned}
P(E) = E[P(E|X)] &= P(E \mid X = 5)P(X = 5) + P(E \mid X = 7)P(X = 7) + P(E \mid X = \text{other}P(\text{other})) \\
&= 1 \cdot \frac{4}{36} + 0\frac{6}{36} + P(E)\frac{26}{36} \\
&= \frac{4}{10} = \frac{2}{5}
\end{aligned}$$

Another way to do it:

$$P(E) = E[P(E \mid N)]$$

Where N is the number of rolls until 5 or 7. $N \sim geom(\frac{10}{36})$

$$\begin{aligned}
P(E) &= \sum_{n=1}^{\infty} P(\text{sum to 5 on nth roll for 1st time} \mid N = \text{first 5 or 7})P(N = n) \\
&= \sum_{n=1}^{\infty} P(\text{sum to 5 on roll n} \mid \text{sum to 5 or 7 on roll n })P(N = n) \\
&= \sum_{n=1}^{\infty} \frac{\frac{4}{36}}{\frac{4}{36} + \frac{6}{36}} P(N = n) \\
&= \sum_{n=1}^{\infty} \frac{4}{10} P(N = n) \\
&= \frac{2}{5} \sum_{n} P(N = n) \\
&= \frac{2}{5} \cdot 1
\end{aligned}$$

because of i.i.d., it doesn't matter whether if its the first time or not
We show here that $E \perp N$.

4.1.2 Markov chains

A stochastic process is a sequence of random variables.

$$\{X_n\} = \text{ stochastic process } = \text{ sequence of R.Vs}$$

Simplest sequence: X_n i.i.d. This has no memory and is stateless.

A Markov chain only considers the most recent thing that happened

MC = markov chain, DTMC = discrete time markov chain.

Definition 4.1.1: Discrete Time Markov Chain

A markov chain is defined as

$$X_{n+1} \mid X_n, X_{n-1}, \dots X_0 \perp X_{n-1}, \dots, X_n$$
$$P(X_{n+1} \mid X_n =, X_{n-1} =, \dots) = P(X_{n+1}|X_n) = P_{ij}$$

In other words, only the previous state matters.
For us, time homogeneous, N doesn't matter

$$P = \|P_{ij}\|$$

This is a transition matrix, where all elements are nonnegative, and the row sum = 1. From state i to state j.

Example 4.1.2 (Gilbert Model of transmission)

G is good State, B is Bad State
We could either stay in each node, or to move between good and bad.
Transition Matrix:

$$P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$$

Example 4.1.3 (4 state markov chain:)

$$X \in \{GG, GB, BG, BB\}$$

$$P(G \mid X) = p_1, p_2, p_3, p_4$$

Going from GG to GG, the probability is p_1. going from GG to GB, that would be $1 - p_1$.
From GB to BG, the probability is p_2. Going from GB to BB, the probability $1 - p_2$
etc.

Example 4.1.4 (Birth Death Markov Chain)

Infinite number of states, but only 2 transmissions. You can either go up and down, or stay in the same state. see textbook for more.

Example 4.1.5 (Inventory Model)

D_i i.i.d. daily demand. If inventory $\leq s$ at day end, order up to S to start next day. (s, S) policy.
State = start of day inventory level

States are from $\{s+1, s+2, i, S-1, S\}$
To stay at state i , $P(D_i = 0)$. to up to S, $P(D_i \geq i - s)$. To go down to $i - 1$, $P(D_i = 1)$, etc.

Chapter 5

5.1 Lecture X: 25 February 2025

$\{X\}$ irreducible $\iff$ there is only one class for all states $(i \leftrightarrow j \forall i, j \in S$

Definition 5.1.1: Period

$$d(i) = \gcd\{N \geq 1 : P_{ii}^n > 0\}$$

i has period $d = d(i)$:

$$P_{ii}^n > 0 \implies \text{d divides n}$$

And all states have the same period.

n is not a multiple of $d \implies P_{ii}^n = 0$, thus aperiodic : $d(i) = 1$

Example 5.1.1 (Period of 2)

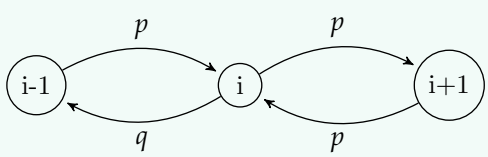

Figure 5.1: period of 2

there are more nodes past $i+1$ with same structure in a birth death cycle. Therefore, you can only get to even nodes or odd nodes depending on i.

5.1.1 Stationary Distribution

X: $X_n \sim \pi$, where π is a row vector for all the probabilities. π is the stationary distribution (equilibrium/invariant) n, if $X_n \sim \pi \implies X_{n+1} \sim \pi$

Example 5.1.2 (Gilbert Model)

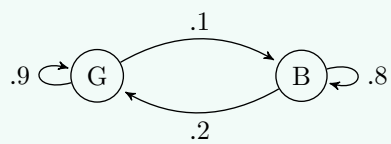

Figure 5.2: Gilbert model

Our intuition states that the stationary dist n should be $\pi = (\frac{2}{3}, \frac{1}{3})$, because we are staying i. $\frac{.2}{.3}$ of the time into G.

$$\pi_G = P(\text{end up in G})) = P(\text{start in G and stay or start in B and move})$$

$$= \pi_g(1-p) + \pi_B(q)$$
$$= p\pi_g = q\pi_B$$

If we solve this for B as well, and use the fact that $\pi = 1$. We get $\pi = (\pi_g, \pi_b) = (\frac{q}{p+q}, \frac{p}{p+q})$
Plugging in $p = .1, q = .2$, we get our original answer.

$$\pi_G = \text{rate of transitions into or out of G}$$

In other words, detailed balance of $\pi_G(1-p) = \pi_B q$

Example 5.1.3 (Packet Transmissions)
Packet example: (G/D/1/2) (Incoming, Outgoing (Deterministic), server #, memory)

$$A_i \sim (p_0, p_1, p_2)$$

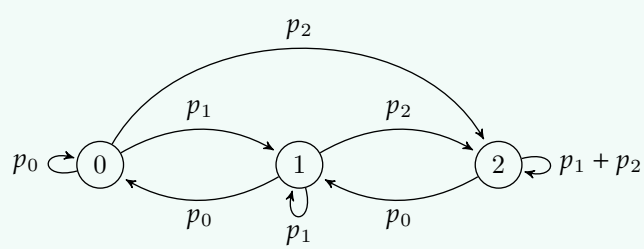

Figure 5.3: packet transmission

What is the stationary dist n?

$$\pi_0 = \pi_0 p_0 + \pi_1 p_0 + \pi_2 \cdot 0$$
$$\pi_1 = \pi_0 p_1 + \pi_1 p_1 + \pi_2 p_0$$
$$\pi_2 = \pi_0 p_2 + p_1 p_2 + \pi_2(p_1 + p_2)$$
$$\pi_1 + \pi_2 + \pi_3 = 1$$

This results in $\pi = (\frac{p_0}{1-p_1} + \frac{p_0(1-p_0)}{1-p_0}, \frac{p_2}{1-p_1})$

Theorem 5.1.1 Stationary Distribution Computation

If π satisfies

1. $\pi_j = \sum_{i \in S} \pi_i P_{ij} \forall j \iff \pi = \pi P$
2. $\sum_{i \in S} \pi_1 = 1$

Then π is a stationary distribution

Proof. Suppose $X_0 \sim \pi$, We need to show that $X_1 \sim \pi$

For $P\{X_1 = j\} = \pi_j$

$$P\{X_1 = j\} = E[P(X_{1=j|X_0})] = \sum_{i \in S} P(X_1 = j \mid X_0 = i) P(X_0 = i) = \sum_{i \in S}^{P_{ij} \cdot \pi_i}$$

Definition 5.1.2: Steady State

The steady state distribuion of $\{X\}$, when it exists, is defined for all j by

$$\lim_{n \to \infty} P(X_n = j \mid X_0 = i) = \lim_{n \to \infty} P_{ij}^n = \text{steady state prob of j}$$

Definition 5.1.3: Long run time average distribution

the long run time distribution of $\{X\}$, if it exists, is defined for all j by

$$\lim_{n \to \infty} \frac{1}{n} \sum_{m=0}^{n} I\{X_m = j \mid X_0 = i\} = \text{l.r. prop time in j}$$

This is weaker than the steady state, basically like an average. counting the number of times we are in state n, resulting in the indicators.

Theorem 5.1.2

For irreducible periodic, positive recurrent MC,

$$P_{ij}^n \to \pi_j, \quad \text{where } \pi \text{ is the stationary distribution}$$

Theorem 5.1.3

For irreducible positive recurrent MC, long run proportion time in j $= \pi_j$, where π is the stationary distribution.